

CMPE 223/242
Hands-On Activity 8

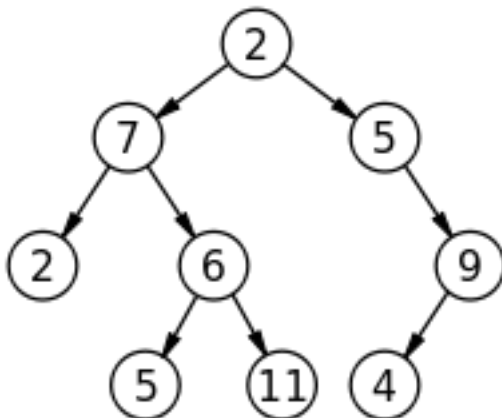
1.

a) Draw the Binary Search Tree whose keys are inserted in the following order:

50 72 96 94 102 26 12 11 9 2 10

b) From the resulting tree in part a, remove the nodes with keys 72, 10, 50 in order.

2. Write the sequence of the nodes visited, during preorder, inorder, and postorder traversal of the tree.

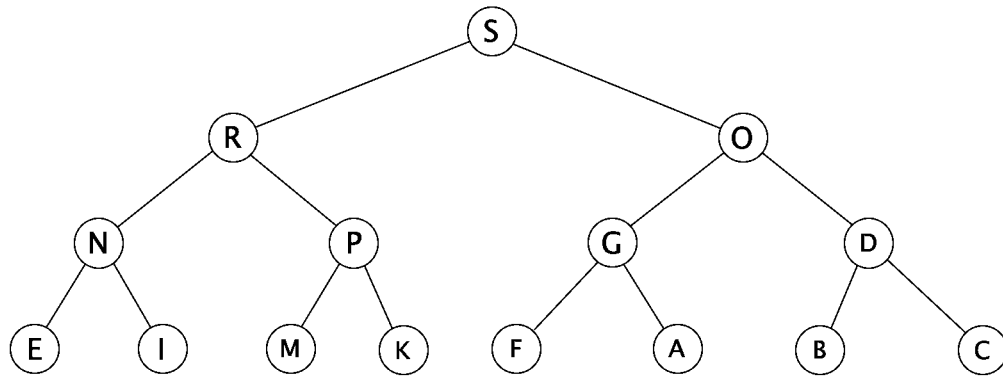


Preorder traversal:

Inorder traversal:

Postorder traversal:

3. a) Insert the key L, T to the max-heap below in that order. Show the resulting heap.



- b) From the resulting heap in part (a), remove a key.

4. Write a **Java method** for a linear-time certification algorithm to check whether an input integer array `pq[]` represents a min-oriented heap. The method should take the integer array `pq[]` and its size as input, and return a `boolean` value with the answer. (Assume that `pq[0]` is unused and the root is at `pq[1]`, the same as in our textbook.)

First, describe your general approach in 1-2 sentences.

Then, write the Java method below.

```
public static boolean isMinHeap( int[] pq, int size)
```

5. Implement a **Java method** that determines whether a given binary search tree (BST) is height-balanced. A tree is considered height-balanced if, for every node, the height difference between its left and right subtrees is no more than one.

Use the following Node class:

```
class Node {  
    int data;  
    Node left;  
    Node right;  
}
```

First, describe your general approach in 1-2 sentences.

Then, write the Java method below (Note: you may use helper function for recursion)

```
public boolean isBalanced(Node root) {
```